Sahaj Bhadja

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Summary

Software engineer with 2+ years building full-stack web applications, backend APIs, SQL-backed workflows, and AI-powered products. Strong in Python, JavaScript/TypeScript, React, FastAPI/Flask, REST APIs, testing, and CI/CD within Agile teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Maintained SQL-backed platform services for 500+ users, tuning indexed access paths, background workers, and reconciliation checks across FastAPI workflows to improve dashboard latency from 1.2 seconds to 350 milliseconds in production.
- Built fault-tolerant LLM job orchestration with explicit state transitions, failure isolation, and CI/CD-backed releases, processing 5,000+ monthly workflows while helping teammates trace issues quickly and deliver stable backend changes.
- Collaborated across backend releases, data fixes, and validation logic, translating stakeholder requests into maintainable service updates and clearer operational debugging paths for teammates supporting recurring platform delivery.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask APIs for an AI grading platform handling 10,000+ daily requests, integrating uploads, rubric generation, grading history, and secure file-processing workflows for instructor-facing educational products.
- Connected multimodal LLM and OCR pipelines to backend services, added concurrent batch processing, and supported CI/CD-backed releases that reduced feedback turnaround by 60% while keeping API behavior predictable during peak usage.
- Used AI coding tools to accelerate implementation, debugging, and prompt iteration, while participating in fast-release workflows that connected frontend state, backend endpoints, and instructor feedback loops across grading features.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL integrations using Python, SQL, AWS Glue, and S3, improving response time by 30% while supporting scalable data collection workflows and maintainable service-to-service integration patterns.
- Maintained 20+ automated test suites and CI pipelines with Jenkins, PyTest, and Mockito, increasing release reliability by 68% and strengthening debugging discipline across multi-environment builds, validation checks, and deployment workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built MERN platform features with REST APIs, WebSockets, and responsive interfaces, supporting synchronized records across modular product workflows while translating business requirements into reliable end-to-end user-facing functionality.
- Automated Docker-based build and test flows with Jenkins, cutting release time by 50% and improving deployment consistency, troubleshooting documentation, and shared development practices across distributed project environments.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash
Frontend: React, Next.js, HTML5, CSS3, TailwindCSS, responsive design, component libraries
Backend & APIs: FastAPI, Flask, Node.js, Express, REST APIs, WebSockets, background workers, PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL
Cloud & Delivery: AWS, Azure, Docker, Jenkins, GitHub Actions, CI/CD, Git, relational modeling, query optimization, LLM integrations, OCR pipelines, prompt workflows, unit testing, code reviews, Agile/Scrum, debugging

Projects

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Built a healthcare workflow platform with TypeScript, MERN, ABDM APIs, Twilio alerts, and Sentry monitoring, reducing administrative workload by 30% through consent-aware patient access, scheduling, and operational dashboards.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-powered SQL workspace across Java and TypeScript services, supporting 10+ databases and accelerating query authoring, schema exploration, and validation through a unified developer interface and Docker-based local workflows.

Linguistix | *Python, AWS, Docker, CI/CD*
- Developed a containerized Python REST API on AWS with automated CI/CD, validation checks, and reproducible releases, achieving 87% classification accuracy while supporting reliable data-processing workflows in production-style environments.